M Naqvi Khan

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EDUCATION

University of Texas at Dallas **Richardson, TX**
BSc in Computer Science (GPA 3.3/4.0) *Graduation Date: December 2021*
Awards: 2x – Competitive Scholarship Recipient

WORK EXPERIENCE

Algorizin Inc. **Remote**
Software Engineer *April 2023 – October 2024*

- Built RESTful APIs in **Spring Boot**, focusing on extensive unit and integration test in **Junit, Selenium** and **Cucumber** resulting in overall code coverage of 85%.
- Implemented **OAuth 2.0** in legacy application using **Spring Security OAuth** for secure API authentication via **AWS API Gateway**, enhancing security and increasing legacy client count by 25%.
- Engineered and scheduled cron jobs to migrate **SQL** database tables, focusing on reducing join operations and improving data normalization, leading to a higher query efficiency.
- Refactored existing **React** components utilizing **Redux** stores, reducing code complexity improving application performance.
- Gained hands-on experience with Agile and **CI/CD** tools (JIRA, GitHub, Jenkins) to streamline development workflows.

SimpleCitizen Inc. **Remote**
Workflow Engineer *May 2022 – April 2023*

- Programmed reusable **Python** scripts for workflows, reducing code redundancy and saving 5 hours weekly in development time.
- Optimized data security by implementing **AES-256-GCM** encryption in **Django**, reducing encryption overhead by 30%.
- Automated case events in **Django**, increasing application process efficiency by 20% via real-time updates to lawyers and clients.
- Developed immigration web application features with **Python**, and **JavaScript**, resulting in a 15% reduction in bounce rate.
- Modified over 200 immigration processing workflows using **XML** and **JSON**, yielding a 30% reduction in case processing time.

Big Data Analytics Lab at UT Dallas **Richardson, TX**
Research Volunteer *January 2020 – May 2020*

Covid Tracking Application

- Worked with a PhD candidate to develop a Covid Tracking data visualization iOS Application using **Swift**.
- Designed layout and statistical animation and utilized **Google Maps API** for map visualization and color-coding regions from low-risk to high-risk.
- Leveraged **JSON** datasets from authoritative sources, resulting in an intuitive application for users to track covid cases by region in Bangladesh.

Data Query Optimization

- Conducted performance research on Redis and Memcached for determining what would be the ideal solution for optimizing data query for a specific API.
- Utilized metrics such as query time, latency, capabilities of caching, in-store memory, and containerization techniques, when presented with large requests to come to optimal conclusion.

ML Pipeline Validation

- Designed and implemented validation frameworks for ML training pipelines to ensure data quality and model reliability on large-scale, unlabeled COVID-19 datasets using Python, TensorFlow, and statistical anomaly detection

Office of Information Technology at UT Dallas **Richardson, TX**
ITSM Student Supervisor *May 2019 – December 2021*

- Led a team of 30+ support analysts, utilizing **mentorship** and **training** to achieve a 20% reduction in customer support time.
- Provided adept troubleshooting for **Windows**, **Mac OS**, and **network connectivity**, ensuring uninterrupted user experiences.
- Efficiently managed **user accounts**, **permissions**, and **group policies** via **Active Directory** to optimize access control.
- Conducted **workshops** for 150+ staff and faculty on against **phishing** and **spam** attacks and lowering incident reports by 20%.
- Maintained high **customer satisfaction** with 90+ weekly tickets and consistently garnered 95% positive survey feedback.

PROJECTS

ML Decision Tree – (Python, Scikit-Learn, Pandas, HTML, CSS, JavaScript)

- Modeled an ML Decision Tree in **Python** (**scikit-learn**, **pandas**) for refined allocation in school tech loaner programs.
- Implemented a user-friendly UI using **HTML**, **CSS**, and **JavaScript** to reduce the time required for users to input information.
- Augmented decision tree model using a dataset of 50,000 entries, resulting in significantly enhanced allocation efficiency.

Fraud Detection Microservice (Capstone, sponsored by Capital One)

- Engineered a microservice using **Java** and **Spring Boot** that reduced anomaly detection time from 15 minutes to under 12 seconds.
- Constructed an OCR using **Tesseract** for processing and a parser for extracting key-values from paystubs with 92% accuracy.
- Deployed rules to **CI/CD** pipelines to test extracted information utilizing **Drools** achieving 95% accuracy in anomaly detection.
- Utilized open-source tools with **Maven** Build, **JUnit** for unit testing, **Git** for version control, incurring $0 development costs.

Workflow Automation – (**Python**, **Selenium**)

- Produced a Python automation script using Selenium for the school's IT office, which enabled lightning-fast logins to work accounts, tools, web applications, and VPN access, reducing a 5-minute task to under 15 seconds.

Pomodoro Timer – (**React**, **JavaScript, jQuery**)

- Engineered a feature-rich web application using React, JavaScript, and jQuery to significantly enhance user productivity through the implementation of the Pomodoro technique.

TECHNICAL SKILLS

Languages: Java, C++, C, C#, Python, HTML, CSS, JavaScript, Typescript, Swift, Objective-C, Verilog, MIPS Assembly, R
Frameworks/Libraries: Spring Boot, Django, React, NodeJS, NextJS, jQuery, Bootstrap, Selenium, PyTorch, Cryptography
Development Tools and Databases: Git, Unix/Linux, Docker, AWS, Jira, Maven, MySQL, PostgreSQL, MongoDB, Memcached

LEADERSHIP

Bangladeshi Student Organization, UT Dallas – *President* *July 2021 – December 2021*

- Constructed organization agendas, providing clear direction and guidance for team members.
- Led and managed a team of 10 committee chairs, overseeing their activities and ensuring smooth coordination.
- Facilitated successful scholarship fundraisers, networking events, and online web-series, resulting in increased donations by 4x the amount of the previous fiscal year and student engagement by approximately 33%.
- Implemented effective digital marketing strategies to promote events and engage with students, resulting in significant increases in donations and student engagement.

  Recognitions:
  Golden Comet Award (Best Student Organization), Best Hospitality Award (International Week)
  Previous Positions Held:
  Webmaster, General Secretary, Vice President